Short questions

|!|BCESB0050000001|!| ★ SI

The enzyme lipase catalyses the breakdown of lipids.

(a) Write a simple word equation to represent the reaction catalysed by lipase. (1 mark)

(b) An experiment was carried out to investigate the effect of temperature on the activity of lipase. The reaction mixtures were maintained at three different temperatures for an hour. The initial pH and final pH of the mixtures were measured and recorded in the table below.

Temperature at which the reaction mixture was kept (°C)	Initial pH of the mixture	Final pH of the mixture
5	7.0	6.8
40	7.0	5.9
70	7.0	7.0

(i) Describe and explain the result of the mixture kept at 40 °C. (3 marks)

(ii) Explain the relative inactivity of lipase at each of the other two temperatures. (2 marks)

##

(a) lipid $\xrightarrow{\text{lipase}}$ glycerol + fatty acids (1)

(b) (i) The pH of the mixture decreased. (1)

Lipase catalyses the breakdown of lipids, forming glycerol and fatty acids. (1)

The formation of fatty acids results in the increase in acidity. (1)

© 2019 Aristo Educational Press Ltd.

(ii) At 5 °C, the kinetic energy of the substrate and enzyme molecules is low, so the chance of forming an enzyme-substrate complex is low. (1)
At 70 °C, the enzymes are denatured by the high temperature and have lost their catalytic function. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Applying concepts	3
(b)(ii)	Applying concepts	2

|!|BCESA0050000002|!| ★★ **STSE**

Mrs Lam and her son Johnny are comparing the cost-effectiveness of two brands of washing powders. Johnny washed some egg-stained clothes with the two different washing powders at 30 °C and at 80 °C respectively. The results are shown in the table below.

	Washing powder	
Temperature (°C)	**Brand A** ($40 / 300 g)	**Brand B** ($60 / 300 g)
30	Egg stain remained	Egg stain disappeared
80	Egg stain disappeared	Egg stain remained

(a) Deduce which of the two brands is a biological washing powder. (3 marks)

(b) Taking into account the effectiveness and cost of the two brands of washing powders, which brand would you recommend Mrs Lam to use? Give a reason to support your answer. (2 marks)

##

© 2019 Aristo Educational Press Ltd.

(a) Brand B is a biological washing powder. (1)
Compared with brand A, brand B removes egg stain at a lower temperature. (1)
However, brand B can no longer remove egg stain at high temperature. This suggests that brand B contains enzymes which denature at high temperature. (1)

(b) Brand B is recommended. (1)
Brand B removes egg stain at a lower temperature. This saves energy despite the higher price. (Or other reasonable answer) (1)

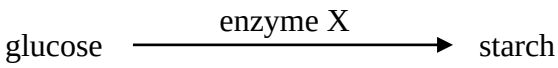
(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Critical thinking	2

|!|BCESB0050000003|!| ★★ SI

Potatoes contain an enzyme X which catalyses the synthesis of starch from glucose:



A student prepared an extract of enzyme X from potato tissue:

- He cut a potato tuber into small cubes.
- He ground the small cubes of potato tissue with distilled water, using a mortar and pestle.
- The resulting mixture is filtered to collect an extract of enzyme X.

(a) Is the reaction catalysed by enzyme X a catabolic reaction or an anabolic reaction? Explain your answer. (2 marks)

(b) What is the purpose of grinding the potato tissue? (1 mark)

(c) Outline how you would conduct an experiment to show that enzyme X is necessary for the conversion of glucose to starch. (4 marks)

© 2019 Aristo Educational Press Ltd.

##

(a) An anabolic reaction (1)

In the reaction, the more complex starch molecules are synthesized from the smaller and simpler glucose molecules. (1)

(b) To break cells open to release the enzymes (1)

(c) Prepare two set-ups:

Add a fixed volume of enzyme X extract and glucose solution to test tube A. (1)

Add a fixed volume of distilled water and glucose solution to test tube B. (1)

Test the contents of the two tubes with iodine solution at the beginning and after one hour. (1)

Iodine solution turns blue-black in the presence of starch. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Practical skills	1
(c)	Practical skills	4

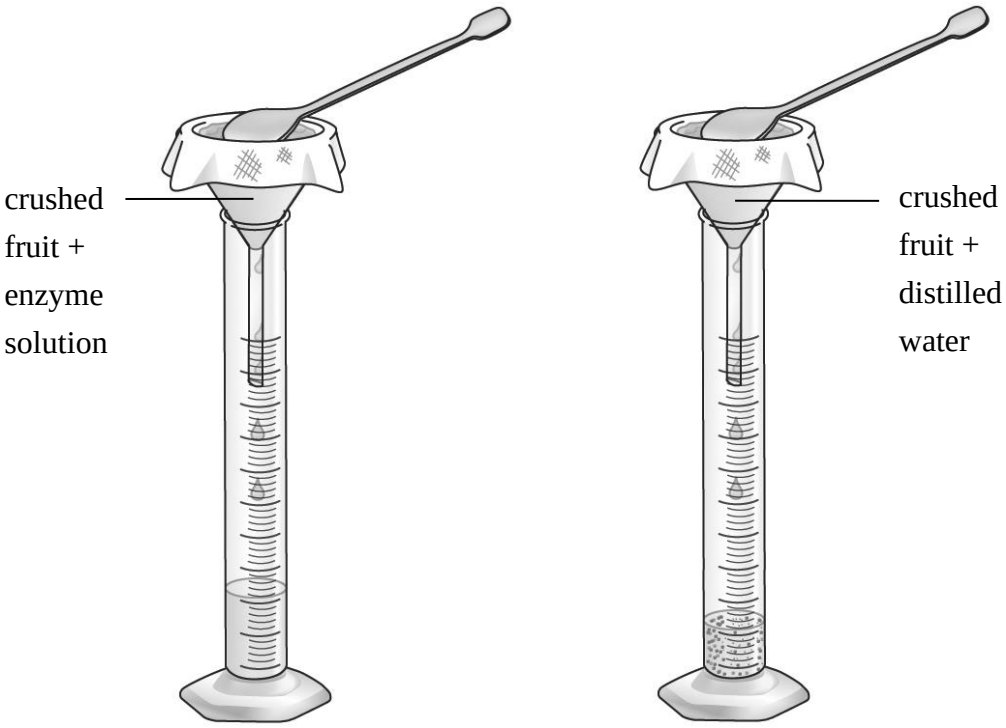
|!|BCESB0050000004|!| ★ **SI**

In an investigation, 20 g of crushed fruit was mixed with 10 cm^3 of an enzyme solution. After 30 minutes, the mixture was filtered to collect juice as shown in Figure I.

Another 20 g of crushed fruit was mixed with 10 cm^3 of distilled water. After 30 minutes, the mixture was filtered to collect juice as shown in Figure II.

© 2019 Aristo Educational Press Ltd.

Figure I ***Figure II***



(a) State ***two*** variables that should be kept constant during the experiment. (2 marks)

(b) Suggest why distilled water was added to the crushed fruit in Figure II. (1 mark)

(c) The enzyme used in Figure I is used commercially in the large scale production of fruit juice. Based on the above results, suggest ***two*** advantages of using the enzyme in fruit juice extraction. (2 marks)

(d) Suggest ***one*** way in which the investigation could be improved to increase the reliability of the results. (1 mark)

© 2019 Aristo Educational Press Ltd.

##

(a) Any ***two***: (2)

Temperature / same amount of stirring / same degree of crushing / same type of fruit

(b) To allow fair comparison with the set-up in Figure I (1)

(c) Using the enzyme, a larger volume of juice can be extracted. (1)

The juice obtained is clearer. (1)

(d) Repeat the experiment for several times and calculate the average values.

OR

Allow more time for enzymes to work on the crushed fruits.

(Or other reasonable answers) (1)

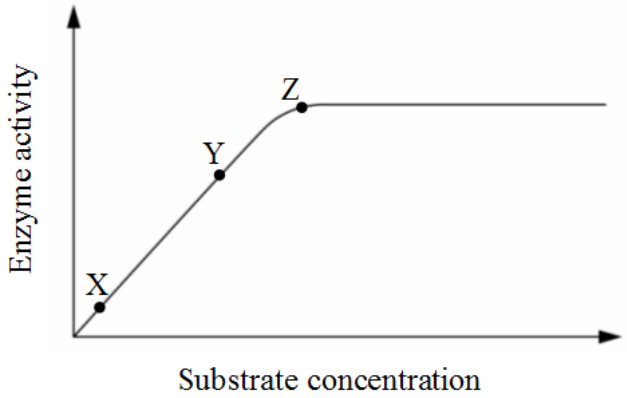
(6 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Analysing data	2
(d)	Practical skills	1

|!|BCESB0050000005|!| ★

The graph below shows the effect of substrate concentration on enzyme activity.



(a) The enzyme concentration was kept constant in the experiment. Identify the factor that was varied by the experimenter. (1 mark)

© 2019 Aristo Educational Press Ltd.

(b) What was the effect of increasing this factor over region XY of the graph? (1 mark)

(c) Suggest a factor limiting the rate of reaction at point Z. (1 mark)

(d) Which letter on the graph represents the situation where

(i) all the active sites of the enzymes are occupied. (1 mark)

(ii) most of the active sites of the enzymes are available for the attachment of substrate molecules. (1 mark)

##

(a) Concentration of substrate (1)

(b) Enzyme activity increased with the increase in substrate concentration. (1)

(c) Concentration of enzymes (1)

(d) (i) Z (1)

(ii) X (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Applying concepts	1
(d)(i)	Analysing data	1
(d)(ii)	Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCESB0050000006|!| ★ **STSE**

Explain each of the following situations in terms of enzymes.

(a) High fevers with a body temperature over 42 ^{o}C are fatal to humans. (2 marks)

(b) Vinegar can be used to preserve food. (2 marks)

##

(a) Important enzymes in body cells are denatured (1)
and are not able to catalyse the biochemical reactions that are essential to sustain life. (1)

(b) Vinegar is acidic. (1)
Some microorganisms cause food to turn bad. Enzyme activity in microorganisms decreases in acidic conditions. (1)
Therefore, microorganisms cannot grow in food preserved with vinegar.

(4 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	2

|!|BCESA0050000007|!| ★★

Diastix papers can be used to diagnose diabetes by detecting glucose in urine. Different concentrations of glucose in urine will give different colour changes on the test end of the paper. The following photograph shows the colour chart for reading the results of the test end of Diastix papers.

© 2019 Aristo Educational Press Ltd.



Source: http://en.wikipedia.org/wiki/File:Ketodiastix.jpg

Two enzymes are present on the test end of a Diastix paper: glucose oxidase and peroxidase. They catalyse the following reactions:

$$\text{glucose + oxygen + water} \xrightarrow{\text{glucose oxidase}} \text{gluconic acid + hydrogen peroxide}$$

$$\text{hydrogen peroxide + potassium iodide} \xrightarrow{\text{peroxidase}} \text{iodine + water}$$

Since potassium iodide is colourless, a blue dye is added to the test end of the Diastix paper for easy observation of a colour change.

(a) Explain why the Diastix paper detects glucose, but not other sugars. (2 marks)

_____

_____

(b) Suggest why peroxidase is needed on the Diastix paper, in addition to glucose oxidase. (2 marks)

_____

_____

(c) Suggest ***one*** advantage of having the two enzymes attached to the Diastix paper, rather than adding them to a urine sample in a test tube. (1 mark)

_____

(d) What treatment should patients receive to control diabetes? (1 mark)

_____

_____

##

© 2019 Aristo Educational Press Ltd.

(a) Glucose oxidase is an enzyme with specific active site. (1)
Only glucose complementary to the active site can combine with it. (1)
(b) It catalyses the reaction to form iodine (1)
which gives a colour change / visible change. (1)
(c) Any ***one*** of the following: (1)
- Test can be done in most situations / not only in laboratory.
- The strip is portable.
- It gives (semi-)quantitative results.

(Or other reasonable answer)
(d) Regular insulin injection (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	2
(c)	Applying concepts	1
(d)	Understanding basic concepts	1

|!|BCESC0050000008|!| **SI**

An experiment was carried out to investigate the effect of pH on the rate of reaction of a digestive enzyme. The results are shown in the table below.

pH	1	2	3	4	5	6	7
Reaction rate (arbitrary units)	10	15	9	6	3	1	0

(a) Suggest in which part of the human digestive system this enzyme would be found. (1 mark)

(b) The experiment was repeated at pH 2 but the enzyme was boiled before use. Suggest how the results would be different and explain why. (2 marks)

© 2019 Aristo Educational Press Ltd.

(c) The experiment was repeated at pH 2 and with mercury ions added. Suggest how the results would be different and explain why. (2 marks)

##

(a) Stomach (1)

(b) There would be no reaction. (1)

This is because high temperature would have caused the denaturation of the enzyme. (1)

(c) The reaction rate would decrease. (1)

This is because mercury is an enzyme inhibitor. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCESC0050000009|!|

(a) State ***three*** properties of enzymes. (3 marks)

(b) Enzymes are widely used in the food industry. Give ***two*** examples and briefly describe how enzymes are used in each case. (4 marks)

##

(a) Any ***three*** of the following: (3)

- Enzymes speed up chemical reactions.
- Enzymes are proteins.

© 2019 Aristo Educational Press Ltd.

- Enzymes are reusable.
- Enzymes are specific in action.
- Some enzyme reactions are reversible.

(Or other reasonable answer)

(b) Any ***two sets*** of the following examples: (4)

(Each example 1 mark, each description 1 mark)

Example	**Description**
Meat tenderizers	Meat tenderizers contain proteases which soften meat by breaking down proteins.
Fruit juice extraction	Pectinases are used to break down the fruit cell wall and release the juice.
Baby food	Enzymes are added to break down complex food molecules into simpler forms for easier absorption by babies.

(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	4

|!|BCESC0050000010|!|

Fill in the blanks to complete the following paragraph. (6 marks)

At low temperatures, most enzymes are usually (a)_____. As the temperature rises, more (b)_____ is supplied to the substrate and enzyme molecules, causing them to move faster. They collide with one another more often, an (c)_____ will be more likely to form. The reaction rate is the highest at the (d)_____ temperature. If the temperature further increases, the shape of enzyme molecules start to change, the enzymes are said to be (e)_____. The loss of function of enzymes at higher temperatures is (f)_____.

##

(a) inactive (1)

(b) kinetic energy (1)

(c) enzyme-substrate complex (1)

© 2019 Aristo Educational Press Ltd.

(d) optimum (1)

(e) denatured (1)

(f) irreversible (1)

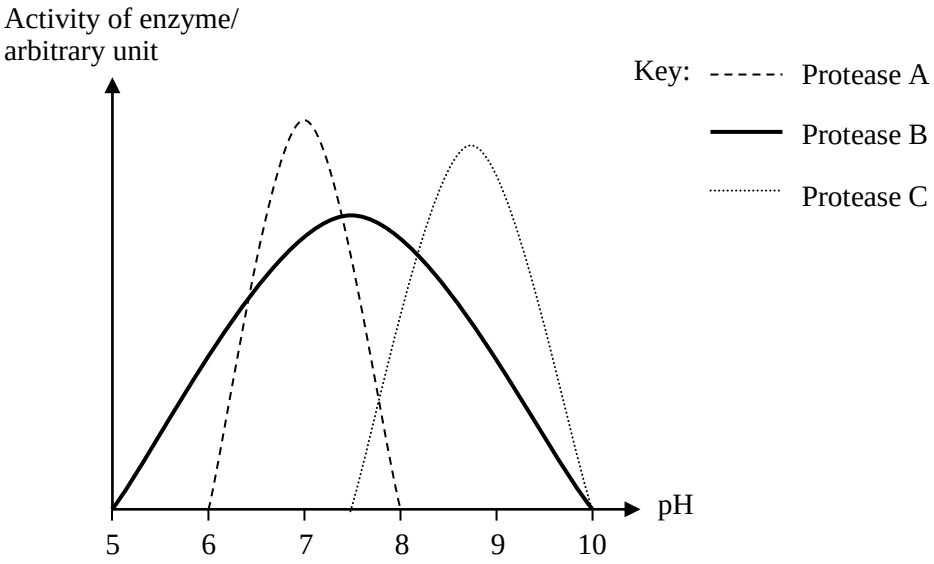
(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESB0050000011|!| ★

The graph below shows the activity of three enzymes over a range of pH values.



Proteases A and B can remove egg stains and protease C can remove blood stains. Suggest which ***two*** enzymes should be used in a washing powder that can be used to remove egg and blood stains. Explain your answer. (3 marks)

##

Proteases B and C should be used. (1)

This is because washing powder is slightly alkaline (with pH 8 to 9). (1)

Both protease B and protease C are active in alkaline medium. (1)

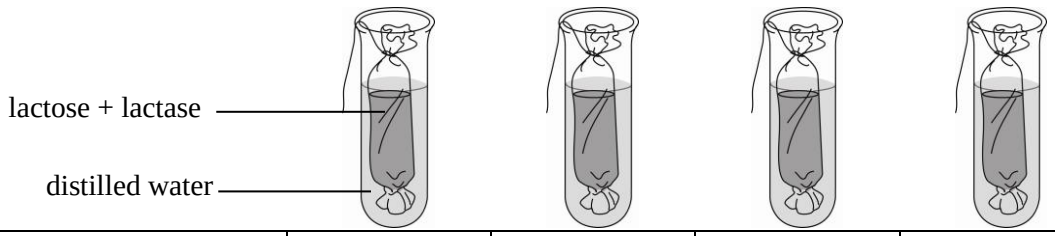
(3 marks)

##

Skills assessed	Marks
Applying concepts	3

|!|BCESB0050000012|!| ★★ SI

Lactase is an enzyme that can digest lactose into simple sugars. An experiment was carried out to investigate the effect of temperature on the activity of lactase. Dialysis tubings were filled with equal volumes of lactose and lactase solution. The tubings were then immersed in water baths of different temperatures.



Temperature (°C)	4	20	38	100
Reaction rate (arbitrary units)	2	9	18	0

(a) Name the simple sugars formed from the digestion of lactose. (1 mark)

(b) Briefly describe how you would determine the reaction rate of lactase in this experiment. (2 marks)

(c) According to the results, describe and explain the effect of temperature on lactase activity. (4 marks)

© 2019 Aristo Educational Press Ltd.

##

(a) Glucose and galactose (1)

(b) Withdraw a certain amount of water surrounding the dialysis tubing at intervals (1)
Carry out Benedict's test on the water withdrawn and measure the depth of red precipitate formed / measure the absorbance of the reaction mixture using a colorimeter (1)

(c) Between 4 ºC and 38 ºC, the rate of enzymatic reaction increases with temperature (1)
because the substrate and enzyme molecules have more kinetic energy at higher temperature and the chance of enzyme-substrate complex formation increases. (1)
From 38 ºC to 100 ºC, the rate of enzymatic reaction decreases to 0 (1)
because the enzyme denatures at high temperature. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Applying concepts	4

© 2019 Aristo Educational Press Ltd.